	Case Name: Railway Bridge (2)	Sector	**Construction (civil)**	
OR-AS Operations Research Applications and Solutions	**OR-AS** Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	■
			Schedule with costs	■
		Risk Analysis	Random simulation	■
			One of nine std. scenarios	
Submitted by	Mieke De Saedeleer		User defined distributions	
Date	2016	Project Control	Automatic tracking	
File Name	C2016-02 Railway Bridge (2)		Tracking based on user input	■

1. Project description

Project authenticity

This is a railway bridge reconstruction project that involves the full lifecycle of replacing or upgrading an existing bridge. The project starts with preliminary studies and site preparation, followed by setting up the construction site. It then proceeds to the demolition of the existing bridge structure, after which new construction activities are carried out. The project likely ends with final inspections and completion of the new railway bridge, ensuring it is ready for operation.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	27
Planned Duration (PD)	229
Budget At Completion (BAC)	962181.55
Renewable Resources	-
Consumable Resources	-

Network topology	
Serial/Parallel (SP)	63%
Activity Distribution (AD)	71%
Length of Arcs (LA)	0%
Topological Float (TF)	82%

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	15	16	2.44
CRI-rho	14	16	2.80
CRI-tau	12	11	2.94

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	65	49	-0.68
SI	7	13	2.13
SSI	11	19	2.77
CRI-r	15	17	2.88
CRI-rho	15	17	3.02
CRI-tau	12	0	2.89

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section "2.2. Risk Analysis". Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	37%	-33%
PV - SPI	53%	11%
PV - SCI	53%	11%
ED - 1	154%	52%
ED - SPI	53%	11%
ED - SCI	53%	11%
ES - 1	28%	-26%
ES - SPI(t)	27%	26%
ES - SCI(t)	27%	26%

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	1%	0%
CPI	1%	1%
SPI	18%	18%
SPI(t)	16%	16%
SCI	18%	18%
SCI(t)	16%	16%
0.8 CPI + 0.2 SPI	14%	14%
0.8 CPI + 0.2 SPI(t)	9%	9%

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 13 tracking periods. The Real Duration and Real Cost mentioned in section "2.3.3. Earned Value Management" are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-6255.59	-48638.21	-17.12	0.99	0.93	0.88	1.00
std dev	4341.73	52983.30	13.28	0.01	0.07	0.10	0.00
final	N/A	N/A	N/A	N/A	N/A	N/A	N/A

2.3.3.2. Time forecasting

PD	229	Real Duration	239	Late/early	4%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	253.92	12.65	6%	-6%
PV - SPI	261.54	19.48	9%	-9%
PV - SCI	264.54	20.89	11%	-11%
ED - 1	254.46	11.42	6%	-6%
ED - SPI	263.69	18.28	10%	-10%
ED - SCI	265.23	19.20	11%	-11%
ES - 1	259.15	13.29	8%	-8%
ES - SPI(t)	276.92	30.55	16%	-16%
ES - SCI(t)	278.54	31.37	17%	-17%

2.3.3.3. Cost forecasting

BAC	962181.55	Real Cost	972341.558	Over/under budget	1%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	968437.22	4341.73	0%	100%
CPI	972503.27	9099.48	1%	100%
SPI	1000920.54	34795.08	3%	100%
SPI(t)	1024258.21	62344.32	6%	100%
SCI	1005522.89	39822.80	4%	100%
SCI(t)	1029119.09	65873.77	6%	100%
0.8 CPI + 0.2 SPI	977607.08	12836.83	1%	100%
0.8 CPI + 0.2 SPI(t)	981055.02	14859.38	1%	100%